

FinRL, FinRobot, and FinGPT: A Layered Portfolio Research Pipeline

Rui Zhao, Qingyue Wang, Ruping Zhang

May 2026

1 Strategy Overview

This project builds a layered portfolio research pipeline that combines three different types of signals. FinRL is used as the quantitative stock-selection engine, FinRobot is used as a point-in-time research confirmation layer, and FinGPT is used as a final news sentiment overlay. The final output is a long-only portfolio that is backtested against QQQ and SPY over the required period from January 1, 2026 to April 30, 2026.

The main research question is whether a structured multi-module pipeline can improve a portfolio decision compared with relying on a single signal source. Price-based and machine-learning signals may rank attractive stocks, but they can miss fundamental weakness or recent negative news. Research-based analysis can add interpretability, but it may be too slow or subjective if it is not converted into a systematic signal. News sentiment can update short-term conviction, but it should not be treated as a standalone buy/sell engine. The pipeline therefore gives each module a narrow role:

- FinRL answers which stocks should enter the weekly candidate basket.
- FinRobot answers whether those selected stocks deserve to be kept, reduced, overweighted, or rejected.
- FinGPT answers whether recent company-specific news should raise or lower conviction.

- The backtest answers whether the final portfolio outperformed QQQ and SPY during the test window.

This design is intentionally conservative. The report does not claim that the system is a fully validated live trading strategy. Instead, it presents a research prototype with transparent intermediate outputs, portfolio weights, and benchmark results.

2 System Architecture

The full system is organized as a sequential signal flow. Each module receives the previous module's output and produces a structured file that can be inspected. The overall architecture is:

Price, fundamental, and news inputs

->

FinRL weekly top-5 stock selection

->

FinRobot fundamental and news confirmation

->

FinGPT no-lookahead sentiment overlay

->

Final portfolio weights

->

Backtest against QQQ and SPY

The FinRL stage produces the initial weekly stock list and equal-weight candidate basket. The FinRobot stage does not create a new universe. Instead, it evaluates the FinRL-selected stocks and produces research multipliers. The FinGPT stage then applies a final news-based conviction signal. The final portfolio combines a fixed QQQ market sleeve with a stock sleeve whose weights are shaped by the layered signals.

The main output files used in this report are:

- FinRL stock selection outputs in `final_stock_selection_2/`.
- FinRobot rebalanced panel: `finrobot2_rebalanced_panel.csv`.
- FinGPT signal file: `fingpt_score_only_signals.csv`.
- Final FRG backtest metrics, benchmark metrics, and dashboard figure in the FRG portfolio backtest folder.

3 Methodology

3.1 Data, Universe, and Preprocessing

The FinRL stock-selection script defines a 20-stock large-cap universe:

AAPL, MSFT, NVDA, META, AMZN, GOOGL, TSLA, AVGO, AMD, NFLX, ADBE, COST,
PEP, QCOM, CSCO, TXN, AMAT, AMGN, INTU, and ISRG.

The selection period is January 1, 2026 to April 30, 2026. The script downloads adjusted close prices beginning on September 1, 2025 and uses an end date of May 1, 2026, because the Yahoo Finance end date is exclusive. The weekly rebalance frequency is W-FRI, and the strategy keeps the top 5 stocks at each rebalance date.

For price-based features, the FinRL script builds:

- 20-day momentum, `mom_20d`;
- 60-day momentum, `mom_60d`;
- 20-day realized volatility, `vol_20d`;
- 60-day rolling drawdown, `drawdown_60d`.

Each feature is transformed into a cross-sectional z-score on each date. Missing price observations are forward-filled using only previously observed prices. Columns or rows that are entirely empty are dropped. This makes the technical feature construction point-in-time and avoids using future returns for ranking.

The fundamental machine-learning data is low-frequency. The implementation uses a quarterly availability schedule. The 2025-09-30 data is assumed available from December 1, 2025 and drives January-February 2026 selections. The 2025-12-31 data is assumed available from March 1, 2026 and drives March-April 2026 selections. This schedule is meant to prevent the January 2026 portfolio from using later quarterly information.

3.2 FinRL Quantitative Selection Layer

The FinRL layer is the first stage of the pipeline. Its job is to convert a broad candidate universe into a weekly top-5 stock basket. The implementation combines machine-learning scores with technical scores:

$$\text{FinRLScore}_{i,t} = 0.60 \times \text{ZMLScore}_{i,t} + 0.40 \times \text{ZTechnicalScore}_{i,t}.$$

The technical score is based on momentum, volatility, and drawdown:

$$\begin{aligned} \text{TechnicalScore}_{i,t} = & 0.40z(\text{mom_20d}) + 0.35z(\text{mom_60d}) \\ & - 0.15z(\text{vol_20d}) + 0.10z(\text{drawdown_60d}). \end{aligned}$$

The positive momentum weights reward stocks with stronger recent returns. The negative volatility weight penalizes unstable stocks. The drawdown term is included as a risk-adjustment feature, so that the ranking is not based only on raw momentum.

The machine-learning layer uses bucket-specific models. The repository output records model families such as random forests, histogram gradient boosting, extra trees, ridge regression, and stacking. The model results are saved in `finrl_ml_model_results.csv`. For example, in the

growth-tech bucket, stacking has the lowest validation MSE for the 2025-09-30 run, while the December 2025 run shows very similar validation MSE values across several growth-tech models. This supports using the ML output as a ranking input rather than treating it as a precise return forecast.

At each weekly rebalance date, stocks are ranked by the final FinRL score and the top 5 are marked as selected. The summary file shows that GOOGL appeared in the weekly top-5 basket 15 times, AMAT 13 times, META 11 times, TSLA 9 times, ISRG 8 times, and TXN and NFLX 7 times each. These counts show that the FinRL stage produced a concentrated technology-oriented candidate set, which is one reason the later research and sentiment filters are important.

3.3 FinRobot Fundamental Confirmation Layer

3.3.1 Role of FinRobot in the Pipeline

In the pipeline, FinRL first selects the weekly top-5 candidate stocks and assigns the initial portfolio weights. FinRobot then acts as a point-in-time research confirmation layer. It does not create a new stock universe or replace the FinRL selection process. Instead, it evaluates the FinRL-selected stocks and decides whether each selected stock should be overweighted, kept unchanged, reduced, or rejected.

This design makes the full pipeline more explainable. FinRL answers which stocks should enter the portfolio, while FinRobot answers how much research conviction each selected stock deserves. A stock may have a strong quantitative signal, but it may still have weak fundamentals, expensive valuation, negative news, or high risk. FinRobot adds an analyst-style validation layer after the quantitative selection step.

The FinRobot module takes the FinRL selection CSV as input. This input contains the weekly selected stocks, their FinRL scores, ranks, and initial weights. FinRobot then generates two main outputs: a multiplier file that contains FinRobot scores, decisions, and research-based multipliers, and a rebalanced panel file that contains the FinRL-selected stocks after FinRobot adjustment. The rebalanced panel is then passed to FinGPT, which applies the final news sentiment overlay on top of

the FinRobot-adjusted weights.

3.3.2 *FinRobot Workflow*

The FinRobot module is implemented as a research-to-signal pipeline. It starts from the FinRL weekly selection file, constructs point-in-time financial and news signals, maps those signals into portfolio multipliers, and produces the rebalanced panel for the next FinGPT stage.

The workflow has three main steps. First, the FinRobot equity analysis script runs the real FinRobot research process for the tickers selected by FinRL. It generates company-level financial analysis files and analyst-style research outputs.

Second, the historical news signal script creates no-lookahead news signals for each date, ticker pair. It uses only news published before the rebalance date and applies company relevance filters to remove unrelated articles. For example, the filter removes false positives such as “Wet AMD,” which refers to a medical condition rather than Advanced Micro Devices.

Third, the FinRobot multiplier script combines the FinRL selected stocks, point-in-time financial data, and historical news signals. It calculates the business quality, valuation, catalyst, and risk scores, then maps the final FinRobot score into a decision and multiplier.

The overall workflow is:

FinRL selected stocks

->

Real FinRobot equity research outputs

->

Point-in-time financial and news signal construction

->

FinRobot score and multiplier

->

After-FinRobot rebalanced weights

->

FinGPT sentiment overlay

3.3.3 Point-in-Time Design and No-Lookahead Control

A key part of the FinRobot implementation is the point-in-time design. Since this project is a trading strategy, the portfolio decision at each rebalance date should only use information that would have been available at that time. This avoids look-ahead bias.

For financial data, the module filters financial statements using `acceptedDate`. For each rebalance date, the script only uses the latest fiscal-year filing whose accepted date is earlier than or equal to the rebalance date:

$$\text{acceptedDate} \leq \text{rebalance date}.$$

If the 2025 annual filing was not available by a specific rebalance date, the script automatically falls back to the most recent available filing, such as 2024A. This prevents early January 2026 portfolio decisions from using financial information released later in January or February.

For catalyst and risk signals, the module uses historical FMP news published before each rebalance date:

$$\text{news publishedDate} \leq \text{rebalance date}.$$

The module also applies ticker, company-name, and company/product-alias filters before using news articles. For ambiguous tickers such as AMD and META, additional exclusion rules are used to remove false positives. This makes the news-based catalyst and risk signals more company-specific.

3.3.4 FinRobot Scoring Framework

For each selected date, ticker pair, FinRobot calculates four component scores:

- `business_quality_score`
- `valuation_score`

- catalyst_score
- risk_score

Each score is normalized between 0 and 1. These four components are then combined into the final finrobot_score.

Business Quality Score. The business quality score measures whether the company has strong operating fundamentals. It combines two parts: an absolute quality score and a peer-relative quality score.

The final business quality score is:

$$\text{BusinessQualityScore} = 0.60 \times \text{AbsoluteQualityScore} + 0.40 \times \text{RelativeQualityScore}.$$

The absolute quality score is:

$$\begin{aligned} \text{AbsoluteQualityScore} = & 0.22 \times \text{GrowthScore} + 0.24 \times \text{MarginScore} \\ & + 0.18 \times \text{EarningsScore} + 0.18 \times \text{CashFlowScore} \\ & + 0.10 \times \text{BalanceSheetScore} + 0.08 \times \text{ROEScore}. \end{aligned}$$

The relative quality score is:

$$\begin{aligned} \text{RelativeQualityScore} = & 0.25 \times \text{GrowthRelativeScore} + 0.25 \times \text{MarginRelativeScore} \\ & + 0.20 \times \text{CashFlowRelativeScore} + 0.15 \times \text{ROERelativeScore} \\ & + 0.15 \times \text{LeverageRelativeScore}. \end{aligned}$$

The reason for combining absolute and relative quality is that FinRobot needs to answer two questions. First, is the company financially strong on its own? Second, among the stocks selected by FinRL in the same week, which companies look stronger relative to the others?

The 60% weight on absolute quality reflects the idea that a company should first have strong

standalone fundamentals. The 40% weight on relative quality reflects that portfolio allocation is also a comparison problem across the selected names in the same rebalance period.

Valuation Score. The valuation score measures whether a selected stock looks attractive or expensive relative to the other FinRL-selected stocks on the same rebalance date.

The formula is:

$$\text{ValuationScore} = 0.40 \times \text{PEPercentileScore} + 0.35 \times \text{EVEBITDAPercentileScore} \\ + 0.25 \times \text{FCFYieldPercentileScore}.$$

For PE ratio and EV/EBITDA, lower values receive higher scores. For free cash flow yield, higher values receive higher scores. Peer-relative valuation is used instead of a single absolute PE threshold because different industries can have different normal valuation levels.

Catalyst Score. The catalyst score measures whether recent point-in-time news provides positive short-term support for a selected stock. It is based only on historical news published before each rebalance date.

The directional score is:

$$\text{DirectionalScore} = \frac{\text{PositiveNews} + 1}{\text{PositiveNews} + \text{NegativeNews} + 2}.$$

The coverage score is:

$$\text{CoverageScore} = \min \left(\frac{\text{NewsCount}}{8}, 1 \right).$$

The final catalyst score is:

$$\text{CatalystScore} = 0.70 \times \text{DirectionalScore} + 0.30 \times \text{CoverageScore}.$$

The smoothing terms +1 and +2 prevent the score from becoming too extreme when there are

only one or two news articles. If no reliable historical news is available for a date, ticker pair, the catalyst score defaults to a neutral value of 0.5.

Risk Score. The risk score captures downside concerns from historical news. A higher risk score means fewer or less severe risk concerns.

The module first computes:

$$\text{RiskIntensity} = \frac{\text{RiskWeightedHits}}{\text{NewsCount}},$$

and:

$$\text{RiskArticleRate} = \frac{\text{RiskNewsCount}}{\text{NewsCount}}.$$

Then it computes the risk penalty:

$$\text{RiskPenalty} = 0.65 \times \text{IntensityPenalty} + 0.35 \times \text{BreadthPenalty}.$$

The final risk score is:

$$\text{RiskScore} = 1.0 - 0.75 \times \text{RiskPenalty}.$$

This revised risk score uses risk density rather than total keyword count. This is important because large companies often have more news articles, so a simple count of risk terms would unfairly punish heavily covered firms.

3.3.5 Composite FinRobot Score

After constructing the four component scores, the module creates the final FinRobot score:

$$\begin{aligned} \text{FinRobotScore} = & 0.45 \times \text{BusinessQualityScore} + 0.25 \times \text{ValuationScore} \\ & + 0.20 \times \text{CatalystScore} + 0.10 \times \text{RiskScore}. \end{aligned}$$

Business quality receives the largest weight because FinRobot is mainly a fundamental research validation layer. Valuation receives the second-largest weight because a strong company may still be unattractive if it is too expensive. Catalyst receives a moderate weight because recent company news can affect short-term price movement, especially in a weekly rebalancing strategy. Risk receives a smaller weight because it is mainly a downside control rather than the primary return signal.

3.3.6 Mapping Scores to Portfolio Multipliers

The final FinRobot score is mapped into a conservative portfolio action:

FinRobot Score	Decision	Multiplier
$\text{score} \geq 0.80$	strong_keep	1.20
$0.60 \leq \text{score} < 0.80$	keep	1.00
$0.45 \leq \text{score} < 0.60$	reduce	0.70
$\text{score} < 0.45$	reject	0.00

The raw after-FinRobot weight is:

$$\text{AfterFinRobotRawWeight}_{i,t} = \text{InitialWeight}_{i,t} \times \text{FinRobotMultiplier}_{i,t}.$$

Then the weights are normalized within each rebalance date:

$$\text{AfterFinRobotWeight}_{i,t} = \frac{\text{AfterFinRobotRawWeight}_{i,t}}{\sum_j \text{AfterFinRobotRawWeight}_{j,t}}.$$

This normalization ensures that the portfolio remains fully invested after FinRobot adjustments. If all raw weights are zero on a rebalance date, the script falls back to the original equal weights to avoid invalid portfolio weights.

3.4 FinGPT-Based No-Lookahead News Sentiment Overlay

3.4.1 Motivation: From Financial News to Trading Signals

The FinGPT module is designed to solve a specific problem in the integrated trading pipeline: unstructured financial news must be converted into a stable numerical signal before it can affect portfolio weights. In the full system, FinRL first ranks stocks using quantitative and price-based information, and FinRobot validates the candidates using fundamental quality, valuation, catalyst, and risk evidence. FinGPT is used as the final short-horizon news overlay. It does not replace the stock-selection model and it does not act as a standalone buy/sell engine. Instead, it adjusts conviction after a stock has already passed the FinRL and FinRobot filters.

This distinction is important for explainability. The role of FinGPT is not to claim that LLM sentiment can directly forecast returns. A more conservative and testable claim is that recent company-specific news can help adjust position size when the news is timely, material, and directionally clear. Therefore, the FinGPT output must be measurable, reproducible, and directly connected to portfolio construction.

3.4.2 No-Lookahead News Window and Input Construction

For each ticker-date pair, the rebalance date is treated as the portfolio decision date. Only news published before that date is allowed to enter the FinGPT module. Let i denote the ticker, t the rebalance date, and K the news-window length in weeks. The no-lookahead window is defined as

$$W_{i,t} = [t - 7K, t - 1].$$

In the implementation, $K = 3$, so each FinGPT score uses only the previous three weeks of company-specific news. The module also applies ticker-specific relevance filters so that broad market headlines do not dominate the signal.

The input to this module is the FinRobot-adjusted rebalanced panel. In the submitted run, the FinGPT batch contains 85 ticker-date observations, covering 11 tickers across 17 rebalance dates

from January 2, 2026 to April 24, 2026. This makes FinGPT a post-research overlay rather than a separate universe-selection model.

3.4.3 Hybrid FinGPT Scoring Design

The first implementation attempted to use the local FinGPT Forecaster as a direct structured-output generator. For each ticker and rebalance date, the model was asked to output three numerical values:

$$s_{i,t} = \text{sentiment score}, \quad c_{i,t} = \text{confidence score}, \quad m_{i,t} = \text{materiality score}.$$

However, direct JSON generation was not stable enough for a batch portfolio pipeline. During testing, the model sometimes produced incomplete JSON objects, malformed outputs, or invalid score patterns such as [1] [1] [1]. A trading system cannot depend on manual cleanup after every malformed LLM response.

To make the pipeline reproducible, FinGPT is implemented as a hybrid no-lookahead scoring system. First, a deterministic scoring layer computes base scores from filtered news:

$$s_{i,t}^0 = \text{base sentiment}, \quad c_{i,t}^0 = \text{base confidence}, \quad m_{i,t}^0 = \text{base materiality}.$$

The base sentiment score is clipped to $[-0.60, 0.60]$, while confidence and materiality are restricted to $[0, 1]$. Positive event terms include earnings beats, analyst upgrades, strong demand, product momentum, and positive guidance. Negative event terms include downgrades, lawsuits, demand weakness, regulatory concerns, delays, and competition risk.

Second, FinGPT is used as a qualitative calibration judge rather than as a direct numerical generator. Given the no-lookahead news and deterministic base scores, FinGPT classifies each score as

$$d \in \{\text{too_high}, \text{too_low}, \text{reasonable}\}.$$

The adjustment function is

$$A(d; \delta) = \begin{cases} -\delta, & d = \text{too_high}, \\ +\delta, & d = \text{too_low}, \\ 0, & d = \text{reasonable}. \end{cases}$$

The calibrated scores are

$$\begin{aligned} s_{i,t} &= \text{clip} \left(s_{i,t}^0 + A(d_{i,t}^s; 0.10), -1, 1 \right), \\ c_{i,t} &= \text{clip} \left(c_{i,t}^0 + A(d_{i,t}^c; 0.05), 0, 1 \right), \\ m_{i,t} &= \text{clip} \left(m_{i,t}^0 + A(d_{i,t}^m; 0.05), 0, 1 \right). \end{aligned}$$

This turns FinGPT into a reliability-oriented calibration layer. It still contributes financial language understanding, but the deterministic layer guarantees that every observation has a valid numerical signal.

3.4.4 Composite News Signal and Portfolio Weight Overlay

The final FinGPT news signal is

$$\text{GPtScore}_{i,t} = s_{i,t} \times c_{i,t} \times m_{i,t}.$$

The score combines direction, confidence, and materiality. A positive value indicates bullish recent news; a negative value indicates bearish recent news; the magnitude indicates how reliable and material the signal is.

The GPT score is converted into a portfolio multiplier:

$$\text{FinGPTMultiplier}_{i,t} = 1 + \lambda \cdot \text{GPtScore}_{i,t},$$

where λ controls the sensitivity of weights to news. The raw final weight is

$$\text{FinalRawWeight}_{i,t} = \text{AfterFinRobotWeight}_{i,t} \times \text{FinGPTMultiplier}_{i,t},$$

and the final portfolio weight is normalized within each rebalance date:

$$\text{FinalWeight}_{i,t} = \frac{\text{FinalRawWeight}_{i,t}}{\sum_j \text{FinalRawWeight}_{j,t}}.$$

The submitted FinGPT signal file contains valid outputs for all 85 ticker-date observations. Among the 85 observations, 59 have positive composite signals and 26 have negative composite signals, with a signal range of $[-0.336, 0.336]$. This shows that the FinGPT overlay is non-degenerate and produces both bullish and bearish adjustments.

3.5 Portfolio Construction and Backtest Rules

The final FRG backtest uses the layered FinRL, FinRobot, and FinGPT signals. The backtest script includes a fixed QQQ market sleeve with weight 60%. The remaining 40% is assigned to the stock basket selected and weighted by the layered pipeline. The backtest uses an initial capital of 1,000,000, a one-trading-day execution lag after signal dates, and a transaction cost parameter of 0.001.

The final weight file shows that QQQ remains the dominant position by design. The stock sleeve rotates mostly across AMAT, GOOGL, TXN, CSCO, AMD, META, NFLX, COST, AMZN, ISRG, and TSLA, although many of these positions are active only in a few weeks or receive very small normalized weights after the overlays.

4 Backtest Results

The final backtest covers January 1, 2026 to April 30, 2026. The main result is shown in Figure 1. The result dashboard includes the growth of one dollar, drawdown, metric snapshot, and target allocation.

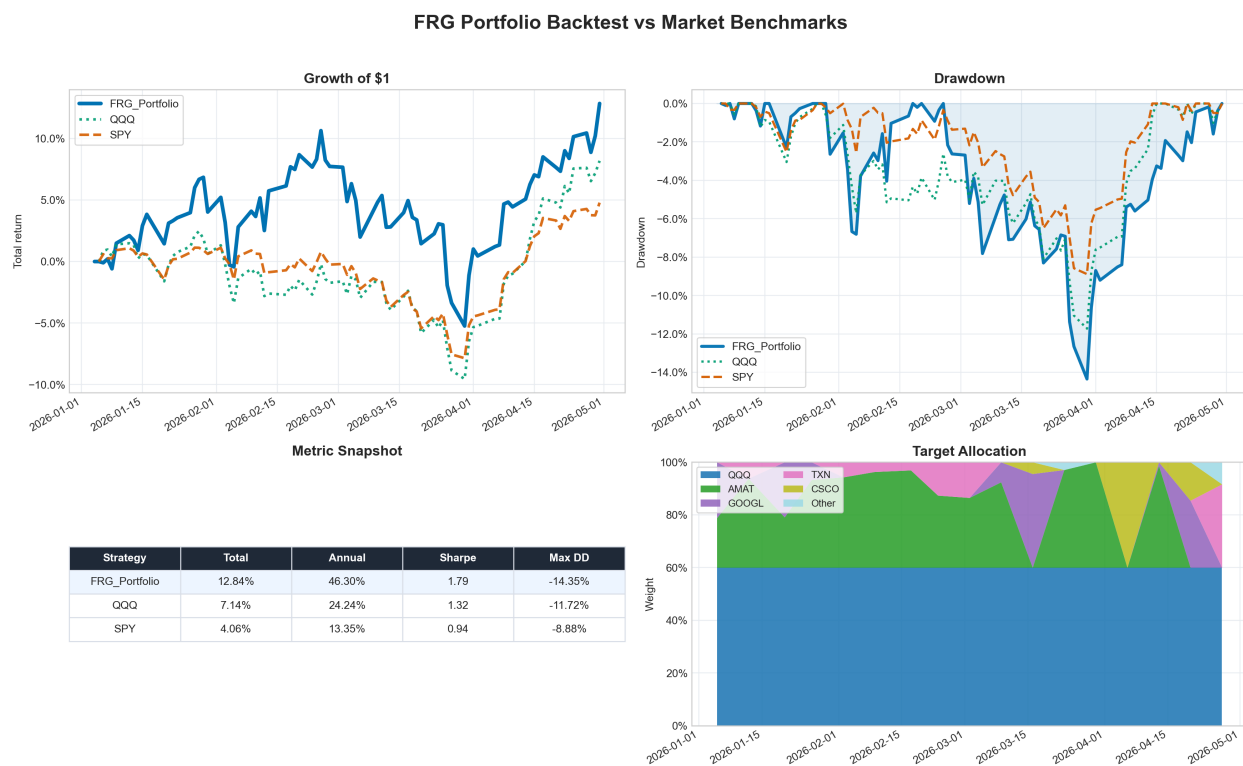


Figure 1. FRG portfolio backtest versus QQQ and SPY. The figure is copied from `finrl/results/frg_portfolio_backtest/frg_portfolio_backtest.png`.

Table 1. Final FRG backtest metrics, January 1, 2026 to April 30, 2026.

Strategy	Total Return	Annual Return	Annual Vol.	Sharpe	Max Drawdown
FRG Portfolio	12.84%	46.30%	25.80%	1.79	-14.35%
QQQ Buy-Hold	7.14%	24.24%	18.41%	1.32	-11.72%
SPY Buy-Hold	4.06%	13.35%	14.16%	0.94	-8.88%

The FRG portfolio outperformed both benchmarks in total return and Sharpe ratio during this

short test period. FRG returned 12.84%, compared with 7.14% for QQQ and 4.06% for SPY. Its Sharpe ratio was 1.79, compared with 1.32 for QQQ and 0.94 for SPY.

The stronger return came with higher downside risk. FRG's maximum drawdown was -14.35%, deeper than QQQ's -11.72% and SPY's -8.88%. Annualized volatility was also higher than both benchmarks. Therefore, the result should be interpreted as a higher-return, higher-risk outcome in this specific backtest window, not as proof of a generally superior live strategy.

5 Module Contribution

Each module contributes a different type of information. FinRL contributes the candidate set and ranking. Without this step, the system would not have a disciplined way to narrow the 20-stock universe into a weekly basket. FinRobot contributes explainability and research discipline by mapping business quality, valuation, catalyst, and risk evidence into portfolio multipliers. FinGPT contributes short-horizon news context and adjusts conviction after the first two filters.

The available result files provide two useful contribution checks. First, the FinRobot backtest output compares a FinRL-only portfolio with a FinRL plus no-lookahead FinRobot portfolio in its own setup. In that file, FinRL-only returns 14.20% with a Sharpe ratio of 1.91, while FinRL plus no-lookahead FinRobot returns 18.91% with a Sharpe ratio of 2.36. This suggests that the FinRobot layer improved the selected basket in that module-specific test.

Second, the final FRG backtest can be compared with the available FinGPT score-only backtest artifact. The FinGPT score-only backtest reports 12.57% total return and a Sharpe ratio of 1.76, while the full FRG backtest reports 12.84% total return and a Sharpe ratio of 1.79. The difference is positive but small. This means the combined system performed slightly better than that score-only artifact, but the evidence does not support claiming a large incremental gain from every module in isolation.

Overall, the combined pipeline is valuable because it makes the decision process traceable. It shows how a stock moves from quantitative selection, to research confirmation, to news-based

conviction adjustment, to final backtest weights. The contribution is therefore both performance-related and interpretability-related.

6 Failure Analysis

The backtest has several important weaknesses.

First, the test window is short. The required period from January 1, 2026 to April 30, 2026 contains only a few months of market data. Annualized return and Sharpe ratio can look large in such a short window, so the annualized metrics should be interpreted cautiously.

Second, the final portfolio is concentrated. The backtest uses a fixed 60% QQQ sleeve, and the 40% stock sleeve frequently concentrates in a small number of names such as AMAT and GOOGL. This helped performance in the submitted period but also makes the strategy sensitive to technology-sector timing.

Third, the final portfolio accepted a larger drawdown than both benchmarks. The strategy beat QQQ and SPY on return and Sharpe, but the -14.35% drawdown shows that the overlays did not fully control downside risk.

Fourth, the news and sentiment layers depend on filtering quality. FinRobot and FinGPT both use historical news signals, relevance filters, keyword logic, and model calibration. These rules make the pipeline reproducible, but they can miss subtle narratives, sarcasm, or events that are already priced in.

Fifth, the repository does not provide a complete robustness suite. Turnover, sensitivity to transaction costs, alternative market sleeves, walk-forward parameter tuning, and longer out-of-sample tests are not fully reported in the available result files. These tests would be needed before treating the method as a validated trading strategy.

7 Conclusion

The final FRG portfolio beat both QQQ and SPY in the submitted backtest from January 1, 2026 to April 30, 2026. It achieved a total return of 12.84% and a Sharpe ratio of 1.79, compared with 7.14% and 1.32 for QQQ and 4.06% and 0.94 for SPY. However, the strategy also had a deeper maximum drawdown of -14.35%, so the result is not a clean risk-reduction story.

The main lesson is that the pipeline is useful as a structured research prototype. FinRL gives the system a quantitative selection backbone, FinRobot adds fundamental and risk confirmation, and FinGPT adds a final news-based conviction overlay. The strongest contribution of the project is not simply that the final return number is higher in this short period, but that the entire decision path can be inspected through intermediate CSV files and final portfolio weights.

Future work should focus on longer evaluation periods, stricter robustness testing, cleaner end-to-end reproducibility, turnover and transaction-cost attribution, and a clearer comparison of each module under exactly the same backtest assumptions. With those additions, the pipeline could move from a promising class project prototype toward a more reliable portfolio research framework.